Speak **L**ike **A** **M**athematician:

indirect proof:

proof by contradiction:

Triangle Inequality Theorem:

Sides, Angles, and the Triangle Inequality

If one SIDE of a triangle is longer than another, then the ANGLE opposite the longer side is ____________________ than the angle opposite the shorter side.

If one ANGLE of a triangle is larger than another, then the SIDE opposite the larger angle is ____________________ than the side opposite the smaller angle.

Triangle Inequality Theorem: the sum of any two side lengths is ____________________ than the third side length. You must check ____________ sums.

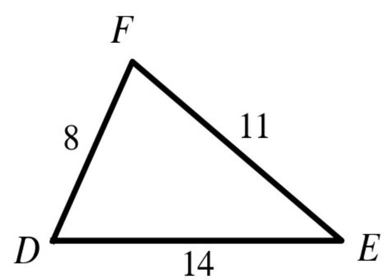
Range for a third side: (difference of the two sides) < third side < (____________ of the two sides).

Example 1 – Order the Angles

In △DEF, DE = 14, EF = 11, and DF = 8. Write the angles in order from smallest to largest.

a) Which angle is opposite each side?

b) Order the angles.

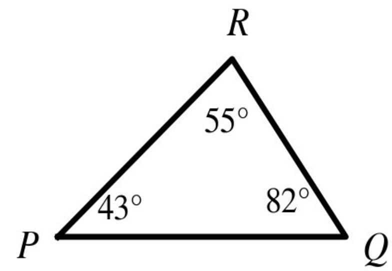


Example 2 – Order the Sides

In △PQR, m∠P = 43°, m∠Q = 82°, and m∠R = 55°. Write the sides in order from smallest to largest.

a) Which side is opposite each angle?

b) Order the sides.



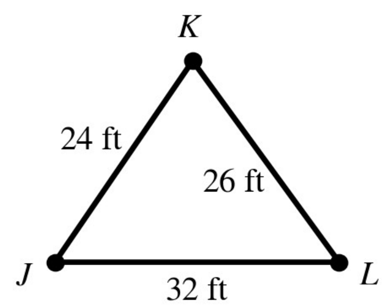
Example 3 – A Stage Prop

A triangular mountain prop has a bottom edge of 32 ft, a left slope of 24 ft, and a right slope of 26 ft. List the angles of △JKL from smallest to largest.

a) Name the side opposite each angle.

b) Order the angles.

c) Check the Triangle Inequality Theorem for this prop.



Example 4 – Can These Be the Sides of a Triangle?

Check all three sums. Explain each answer.

a) 7, 10, 19

b) 2.3, 3.1, 4.6

c) $n + 6$, $n^2 - 1$, $3n$ when $n = 4$

Example 5 – More Practice

a) 6.2, 7, 9

b) 8, 13, 21

c) $t - 2$, $4t$, $t^2 + 1$ when $t = 4$

Example 6 – The Range for a Third Side

Two sides of a triangle measure 8 in. and 13 in.

a) Write the two inequalities the third side x must satisfy.

b) Write the range as one compound inequality.

Example 7 – Writing an Indirect Proof

An indirect proof has three moves: assume the opposite, reason until something impossible happens, and conclude that the assumption was wrong.

Prove: A triangle cannot have two right angles.

Step 1) Assume the opposite of what you want to prove.

Step 2) Reason from the assumption until you reach a contradiction.

Step 3) State the conclusion.

Example 8 – Set Up an Indirect Proof

Write only the FIRST step — the assumption — for each statement.

a) Prove: In $\triangle ABC$, if $AB \neq AC$, then $\angle C \ncong \angle B$.

b) Prove: A triangle cannot have two obtuse angles.

c) Prove: If x^2 is odd, then x is odd.